Define Problem Statements Template

Date	5 July 2026
Team ID	Batch 2023-27
Project Name	Rising Waters – ML-Based Flood Prediction System
Maximum Marks	3 Marks

Step 1: Team Members

S.No	Team Member Name	Role
1	Juturu Haseena	Team Lead
2	Pinjari Nasreen	Member
3	Gyansi Pravallika	Member
4	Lakshmi Sudha	Member
5	Pacharapalli Jyothika	Member

Problem Statement Definition

A clear problem statement focuses the team on the core challenge and guides the design of the solution.

Attribute	Details
Problem Statement	Floods are among the most devastating natural disasters, claiming thousands of lives and displacing millions each year. Conventional forecasting methods lack ML-powered real-time predictive intelligence, leaving authorities with insufficient lead time to respond.
Who is affected?	Disaster management officers, meteorologists, local authorities, and flood-affected communities.

Attribute	Details
What is the gap?	No accessible, ML-powered web application that accepts meteorological inputs and instantly classifies flood risk in real time.
Desired Outcome	A Flask web application using Random Forest (95.65% accuracy) that accepts 10 meteorological features and outputs a binary Flood / No Flood prediction with emergency response guidance.
How Might We?	How might we provide disaster management teams with a reliable, real-time ML flood prediction system that enables timely evacuations and saves lives?